

Decoding Mindfulness With Multivariate Predictive Models

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ABSTRACT

Identifying the brain mechanisms that underlie the salutary effects of mindfulness meditation and related practices is a critical goal of contemplative neuroscience. Here, we suggest that the use of multivariate predictive models represents a promising and powerful methodology that could be better leveraged to pursue this goal. This approach incorporates key principles of multivariate decoding, predictive classification, and model-based analyses, all of which represent a strong departure from conventional brain mapping approaches. We highlight 2 such research strategies—state induction and neuromarker identification—and provide illustrative examples of how these approaches have been used to examine central questions in mindfulness, such as the distinction between internally directed focused attention and mind wandering and the effects of mindfulness interventions on somatic pain and drug-related cravings. We conclude by discussing important issues to be addressed with future research, including key tradeoffs between using a personalized versus population-based approach to predictive modeling.

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One of the central assumptions in mindfulness research is that practicing mindfulness produces beneficial effects for psychological functioning and well-being by changing the brain (1). Much of mindfulness-focused cognitive neuroscience research has been geared toward discovering key neural mechanisms associated with training and practices that can be linked with psychological and behavioral outcomes. However, we still know surprisingly little about these mechanisms. In this review, we suggest that a potential reason for this gap in understanding is related to limitations regarding the way that brain-mind-behavior linkages have been traditionally assessed, particularly in neuroimaging studies (e.g., functional magnetic resonance imaging [fMRI], positron emission tomography, electroencephalography, magnetoencephalography, functional near-infrared spectroscopy). Specifically, most studies involve univariate statistical analyses focused on identifying brain-mind or brain-behavior associations.

Here, we suggest that a shift toward multivariate decoding approaches may provide a more comprehensive assessment of brain activity related to a mindfulness-related state or effect, with the goal of providing quantitative and falsifiable predictions rather than testing associations with unknown and sometimes arbitrarily small effect sizes. Previous tutorial reviews have discussed key features of multivariate decoding and predictive modeling approaches (2–4). For interested readers who are less familiar with these approaches, in the [Supplement](#), we provide an extended discussion of the principles of multivariate analyses, predictive classification and regression, and model-based approaches, which we believe yield key advantages for mindfulness research.

In the sections that follow, we provide illustrative examples of how multivariate decoding and predictive modeling

approaches have been utilized in early studies to understand mindfulness states and to reveal neuromarkers of processes like pain and food/drug craving that may be particularly useful as brain targets for mindfulness interventions. Following this review, we briefly discuss key methodological issues and promising strategies for how these techniques can be deployed to advance our understanding of mindfulness.

USING MULTIVARIATE PREDICTIVE MODELS TO IDENTIFY MINDFULNESS STATES

The advantages of multivariate predictive modeling are becoming more widely recognized and applied in cognitive neuroscience research. However, its use in studies of mindfulness meditation and related contemplative practices remains relatively limited. It is worth noting that the construct of mindfulness itself is complex and still subject to debate (5), because it can variously refer not only to a set of meditation practices but also a trait, cognitive process or skill, type of intervention, or mental state (6). Here, we discuss work aimed at distinguishing mindfulness states from mindfulness as a trait-level variable.

Most of the previous work involving predictive modeling has used electroencephalography or magnetoencephalography methods and has primarily focused on topics that are more in the domain of machine learning, such as issues related to electroencephalography feature selection or classification techniques (7,8). Similarly, the classification questions have been somewhat first-order trait and state questions, such as distinguishing between expert versus novice meditators or between meditation versus rest states, although one recent study examined the effect of mindfulness training on classification of these states (7), while another demonstrated that

mindfulness expertise could be classified in a state-independent manner (9).

One challenge of using multivariate predictive models in mindfulness research is that mindfulness states typically involve internally directed mental processes and cognitive phenomena, which makes them more difficult to measure and analyze. In contrast, standard cognitive neuroscience designs and multivariate modeling approaches often rely on tight experimental control and external markers (stimuli, responses, etc.). Thus, a promising research strategy for utilizing multivariate modeling to decode mindfulness is to bring internally directed cognition under experimental control, via a state-induction approach (10). The basic concept is to induce different mental states in an individual so that the distributed pattern of brain activity associated with each of those states can be identified and distinguished, using tools such as multivariate pattern analysis classification. For example, such multivariate pattern analysis approaches have been used to track the internal focus of attention to information held in mind (11–13). Likewise, in studies of working memory, participants can be directed not only as to what information they should focus on but also as to how they should focus on it. Such work has demonstrated dissociable neural processes associated with rehearsal [e.g., refreshing vs. elaboration (14)] and removal of information from working memory [e.g., replacing, suppressing, and clearing (15,16)].

The key advantage of the induction strategy is that after different mental states have been induced and their neural signatures captured using multivariate pattern analysis-based classifiers, it then becomes possible to apply these classifiers to new contexts for which the experimental control is reduced or not possible, using them as an indicator or readout of the mental states present during these contexts and moreover how they might fluctuate across time. These 2 steps represent the essence of the state-induction approach to decoding mental states: 1) capture activity patterns associated with the induction of different mental states in one phase of the experiment (typically using a localizer task to produce these specific training data) and 2) decode those activity patterns in a separate phase of the experiment to identify and track those mental states across different contexts and over time (Figure 1).

This state-induction approach may be a particularly promising strategy for investigating mindfulness, because it can provide readouts of fluctuating mental states that do not rely on the practitioner having to report which state they are in. For example, in Hasenkamp *et al.* (17) and other related studies (18), individuals likely fluctuate often between different mental states during meditation, and explicit reporting of these fluctuations might have influenced the very states that were reported. Thus, it would be advantageous to develop neural decoders that could discreetly track these state transitions. The idea is to use distributed patterns of brain activity as a proxy for the engagement of distinct cognitive operations that are presumed to be involved when mindfulness is being practiced (e.g., focused attention to the breath vs. mind wandering). Some early studies have used state induction to distinguish mindfulness from mind wandering (19,20), and one study used connectivity-based decoding, instead of activation-based measures, to distinguish between focused

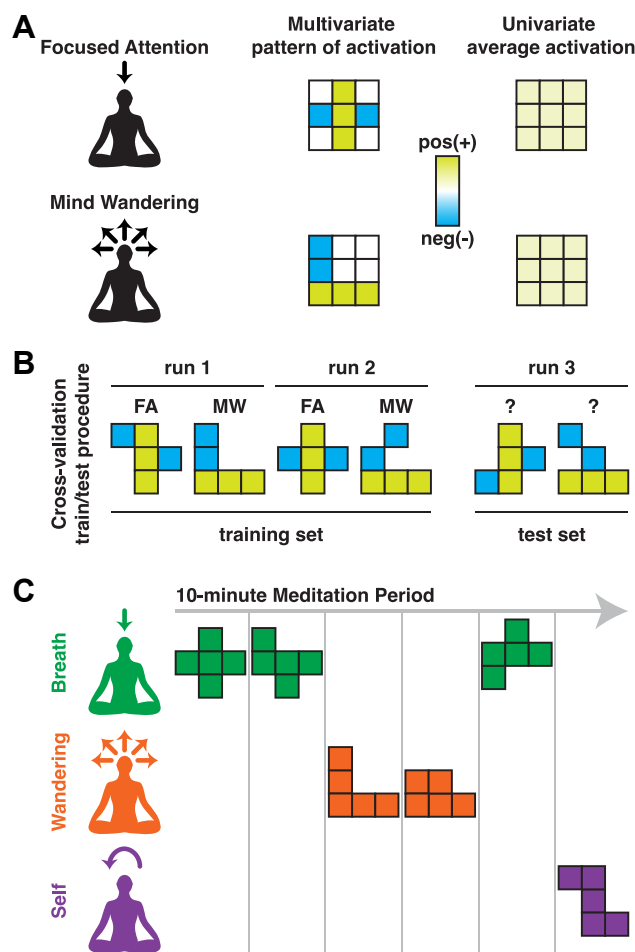


Figure 1. (A) Observing the multivariate pattern of activation across a set of voxels (different colored squares indicating activation intensity) in a brain region of interest allows us to distinguish mindfulness states of focused attention (FA) and mind wandering (MW), whereas univariate averaging of activity in this region does not (right column). (B) An example cross-validation procedure that trains a model on labeled data of (noisy) FA and MW trials from runs 1 and 2 and tests the model on unlabeled data from run 3 to predict whether they most resemble the FA or MW condition, i.e., to predict whether the participant more likely focused or wandered. (C) Stylized recreation of data from a meditator in Weng *et al.* (22) in which multivariate pattern analysis classifiers trained on functional magnetic resonance imaging data from an internal attention localizer task (breath-focused attention, mind wandering, and self-referential processing) decode the mental state of that participant every 1 second throughout a separate 10-minute breath meditation in the scanner.

attention and open monitoring meditation practices (21). However, the 2-step induction approach has been used only rarely.

Weng *et al.* (22) used the 2-step induction strategy with fMRI activation patterns to investigate focused attention meditation. First, an internal attention localizer task was used to train multivariate predictive models to classify and distinguish brain activity patterns associated with 3 meditation-related mental states (attention to breath, attention to body, attention to sound) and 2 additional states (mind wandering and self-referential processing) induced via instructional cues

repeatedly during short, intermixed blocks. Then the trained model was applied to a separate set of fMRI data recorded during a 10-minute period of meditation, in which practitioners attempted to maintain attention to the breath, but no other cues were provided and no responses were required.

Experienced meditators and age- and gender-matched novice control participants were recruited, and multivariate predictive models were trained separately for each participant. Cross-validation testing of the training data from the localizer task revealed successful classification of these 5 mental states in most participants (87%, including all experienced meditators). The important voxels that contributed to accurate classification for each mental state were distributed across many areas of the brain and tended to be unique for each participant. Each participant's model was then applied to data from a separate 10-minute meditation session. The engagement of the 3 most relevant states were tracked from moment to moment, and each time point was classified as reflecting one of these states. Each participant's meditation session was characterized by a state transition graph that showed, for example, breath-focused attention for 20 seconds, followed by mind wandering for 24 seconds, followed by a brief 6-second bout of self-referential processing, and then back to focusing on the breath for 10 seconds, and so on (Figure 1C).

Across participants, the 10-minute meditation period was characterized by greater attention to the breath versus the other mental states, with a greater proportion of overall time and a longer duration for each breath-focused event. This study was an initial proof of concept that internal attentional states could be tracked during meditation, thereby externalizing a key component of the practice that involves the direction of internal attention. Additional work is needed to replicate this approach in different samples of participants, and these neural measures need to be further validated in relation to mindfulness training effects and in relation to participant subjective and behavioral responses of what mental states are present and when during meditation.

As one recent example of such work, Zuo *et al.* (23) adopted the 2-step state-induction approach to carefully distinguish interoceptive attention (to the breath) from exteroceptive attention (to dynamic visual stimuli) using tightly matched experimental localizer conditions. A multivariate predictive model was developed from the localizer conditions and then applied to an extended period in which participants engaged in sustained interoceptive attention to the breath in scanning sessions conducted both before and after mindfulness training or a no-intervention control period. The key finding was that a multivariate predictive model, with key features associated with components of the brain's interoceptive network (insula, posterior cingulate cortex), classified participants receiving mindfulness training as exhibiting an increase in the proportion of time spent in the interoceptive attention state, particularly during the first half of the task period, whereas no such pre-post differences were observed in the control group.

USING MULTIVARIATE PREDICTIVE MODELS FOR NEUROMARKER IDENTIFICATION

In addition to elucidating the neural correlates of mindfulness states directly, multivariate predictive models can be used to

examine the impact of mindfulness practices and training in modulating other psychological states and processes. This aspect of mindfulness research also has strong clinical implications, particularly for problematic states, such as anxiety, pain, and drug craving. In such contexts, the multivariate predictive modeling approach is often referred to as neuro-marker identification. As in the state-induction approach, the key research strategy is also a 2-step process, in which a predictive model is first developed and validated for a psychological process of interest (i.e., a construct), and then after such validation occurs, the neuromarker is utilized in a predictive manner to investigate the effects of various interventions. However, for neuromarker identification, initial validation is the most critical step, and it relies on demonstrating that the predictive model has very high sensitivity, specificity, and generalizability for the psychological construct of interest before it should be utilized in an intervention context.

The example of pain is quite useful to illustrate this research strategy and how it can be applied to investigate mindfulness interventions. Here, the neuromarker is termed the neurological pain signature (NPS), and refers to multivariate predictive model derived from fMRI activation data and optimized to predict the intensity of pain (24) (Figure 2). The NPS was trained to track self-reported pain intensity to varying levels of painful heat, illustrating that such models can be used to predict continuous outcomes like pain in addition to categorical outcomes. Importantly, the NPS involves a widely distributed pattern of brain activity but includes many salient components of the well-established neural pain matrix (anterior cingulate cortex, insula, somatosensory cortex, thalamus). The NPS is arguably the most rigorously validated fMRI-based neuromarker to date in that it has been tested in approximately 50 independent study cohorts under varying conditions, providing a growing database of information regarding its sensitivity and specificity to pain, as well as generalizability across different types of pain, situational contexts, and populations (3,25). Results indicate that 95% of individual participants exhibit a positive pain-related response in the NPS, with an average effect size of $d = 2.32$ (according to standard conventions, $d \geq 1.0$ refers to a very large effect). Likewise, the accumulated data on the NPS have revealed the structure of specificity and generalizability of its response profile: the NPS is highly sensitive to multiple types of evoked somatic pain (e.g., heat, shock, pressure, gastric) but is not sensitive to other similar types of stimuli, including non-noxious warm stimuli, threat cues, aversive images, or observed pain.

The established validity of the NPS as a neuromarker of somatic pain highlights its potential utility for examining whether mindfulness-related interventions that may alter specific aspects of pain neurophysiology are reflected in the NPS. Moreover, the effects of mindfulness interventions can be meaningfully compared against other interventions in terms of NPS impact. For example, the NPS is strongly affected by the opiate remifentanyl but remarkably insensitive to placebo treatments, even those that produce large subjective analgesic effects (26). Likewise, it is also insensitive to established psychological interventions, such as cognitive reappraisal (27).

So how do mindfulness-related interventions impact the NPS relative to these other treatments? One study employed

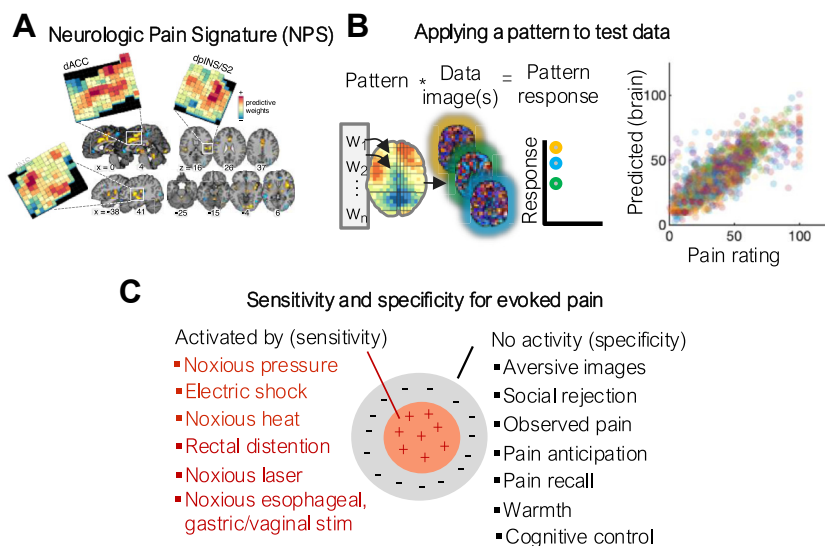


Figure 2. Developing and validating a population-level neuromarker for pain. **(A)** The NPS was trained to predict somatic pain intensity within and across participants. It includes positive pain-predictive weights within regions known to encode and play causal roles in nociceptive pain and negative weights in areas in which more activity predicts less pain. **(B)** The NPS and related population-level patterns can be applied to data from new participants. The pattern is a set of weights applied to a test image to yield a single pattern response value. The right panel shows a plot of reported pain (x-axis) vs. NPS responses (predicted pain) for different levels of painful heat in an independent test sample. **(C)** A current picture of some of the basic affective conditions that the NPS is sensitive to (red) and insensitive to (i.e., specific against; gray) [updated from Woo *et al.* (25)]. dACC, dorsal anterior cingulate cortex; dpINS, dorsal posterior insula; S2, secondary somatosensory cortex.

brief training in mindful acceptance, with instructions to attend to and accept all stimuli as interesting sensations. This study found that even such a brief manipulation reduced NPS responses to a greater degree than 95% of the other interventions tested and with effect sizes comparable to pain modulation achieved through classical conditioning (28). Another study utilized the NPS to examine the effects of a well-established 8-week mindfulness intervention, mindfulness-based stress reduction (MBSR), on their response to evoked pain, relative to both a matched active control intervention (health education program) and to a waitlist group (29). The MBSR group showed significant reductions in the NPS response to painful heat following the mindfulness intervention relative to their preintervention baseline. Furthermore, the NPS response in the MBSR group was also reduced relative to both control groups in the postintervention assessment. These findings suggest that the MBSR intervention can selectively reduce basic nociceptive pain responses, although interestingly there was a nonspecific decrease in subjective pain responses in both the MBSR and health education program groups. Thus, the MBSR-specific effect on the neural response to pain may be partially masked by placebo effects in subjective pain.

The NPS studies provide an example of how neuromarkers of affective processes can be applied to compare mindfulness with other interventions and demonstrate that training in mindfulness, potentially even through brief interventions, can be effective. The findings also raise new questions regarding the potential mechanisms of such intervention effects and how the active ingredients of mindfulness can be optimized through specific types of training. The potential utility of neuromarkers for examining mindfulness effects is not limited to the example of somatic pain. Other neuromarkers have been developed for observed pain, negative affect, reward, and pleasure (30–33). In particular, a neuromarker called the neural craving signature (NCS), comprising a multivariate pattern of activity involving regions previously established to be involved in reward and

addiction (ventromedial prefrontal cortex, ventral striatum, dorsal anterior cingulate), provides a predictive model of food and drug craving states (34). Like the NPS for somatic pain, the NCS was established to have high validity in terms of sensitivity, specificity, and generalizability through a similar validation process, including tests in healthy control individuals and individuals with substance use disorder. After being validated, the NCS has also more recently begun to be tested as a target of various interventions, including those involving mindful acceptance, i.e., creating psychological distance from cues and attending to craving-related sensations without elaborating on the desire to consume. With such interventions, the NCS showed a reduced response relative to a psychological control condition, suggesting the potential for additional investigation regarding longer-term changes (see Figure 3).

IMPORTANT CONSIDERATIONS FOR FUTURE MINDFULNESS RESEARCH

The examples described above demonstrate how both the state induction and neuromarker identification approaches provide new insights into mental states directly associated with mindfulness (e.g., breath-focused meditation and interoceptive attention) as well as other psychological states that may be directly impacted by mindfulness (e.g., pain and craving). Moreover, the use of multivariate predictive models suggests the possibility of building a whole neural dictionary of inner states relevant to mindfulness so that their commonalities, differences, and interactive properties can be explored and characterized.

However, an important practical question is how to best build this neural dictionary of psychological and mental states. Should it be defined uniquely for each individual or constructed at the group or population level? The answer to this question is not clear. In the work that has used a state induction approach (22,23), the focus was on individualized assessment. That is, the target brain states were identified separately for each

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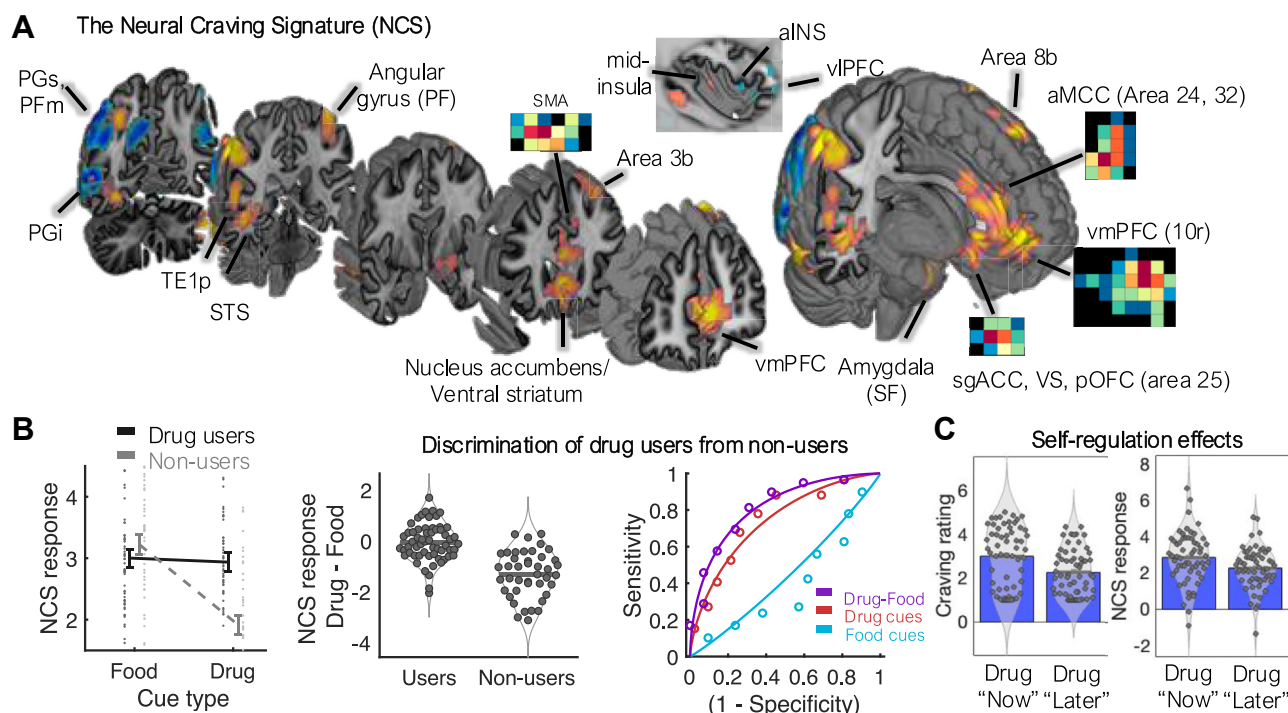


Figure 3. The NCS. **(A)** Areas with significant weights, indicating stable contributions to prediction of craving. **(B)** The NCS responds to food cues in both drug users (substance use disorders for cocaine, alcohol, nicotine) and control individuals but shows strong responses to drug cues only in users. The drug cue response, or drug-food response, classifies users and nonusers with >80% accuracy. **(C)** In both users and nonusers, both drug craving and NCS responses are reduced when participants apply a mindful acceptance and distancing-related strategy ("Later" condition) compared with focusing on appetitive aspects of the cues and consumption ("Now"). aINS, anterior insula; aMCC, anterior midcingulate cortex; PF, parietal field; PFm, parietal field medial; pOFC, posterior orbitofrontal cortex; SF, superior frontal gyrus; sgACC, subgenual anterior cingulate cortex; SMA, supplementary motor area; STS, superior temporal sulcus; vIPFC, ventrolateral prefrontal cortex; vmPFC, ventromedial prefrontal cortex; VS, ventral striatum.

participant and thus could be unique and idiosyncratic. Such an approach is well suited for an intersectional perspective when highly diverse samples are employed and the focus is on developing personalized neural metrics of meditation practice and skill (35). Nevertheless, there are clear drawbacks and downsides to this type of approach. To build individualized classifiers, it is necessary to collect substantial training data from an independent localizer task, which requires additional experimental costs in terms of time and money.

Conversely, a key advantage of the personalized approach is that it may increase sensitivity to detect idiosyncratic mental states present within individuals, particularly if the brain representation for such states varies substantially across individuals (36). This advantage may be especially important for mental states that are already suspected to exhibit strong individual differences, such as those associated with higher cognitive functions. As an example, Figure 4 shows classification results for 3 different working memory operations: maintain, replace, and suppress (37). Classification performance was superior for within-subject analysis (training and testing on data from the same individual) as opposed to training on data from other participants in the study (between participants) or from other participants from a different experiment altogether (between experiments). This result suggests that there may be value in spending the effort and cost to capture individualized predictive models to achieve greater

accuracy in detecting higher cognitive mental states. This approach fits well with the increasing focus on precision neuroscience and functional mapping (38), which involves collecting detailed data from multiple sessions with the same participant to create more accurate, personalized models of brain organization and function.

Given the idiosyncratic nature of mindfulness training and the progression of skill development, there are good reasons to think that mental states related to mindfulness likely exhibit very high between-subject variability. Moreover, within this domain, individualized models may provide the potential to address a range of important research questions. For example, one of the most important questions in mindfulness research is related to neurophenomenology—linking different brain states to different properties of subjective experience (39,40). Current work is focusing on detailed computational approaches to better formalize this process, and multivariate predictive models are likely to be an essential tool in this endeavor (41–43). However, for detailed and precise mappings of the relationship between brain states and phenomenological ones, it is likely that this work will need to be conducted on an individualized basis. Moreover, one of the long-term goals of multivariate predictive modeling research is to use decoded brain states presumed to be associated with beneficial mindfulness states as targets for adaptive neurofeedback training (i.e., to enhance the likelihood or intensity of visiting and

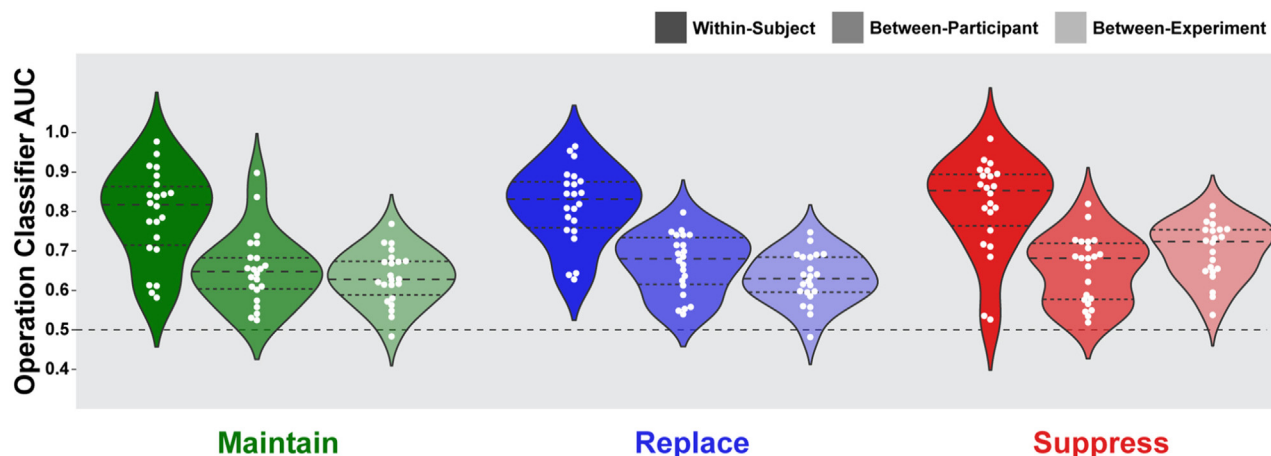


Figure 4. Multivariate pattern analysis decoding of working memory operations. Whole-brain functional magnetic resonance imaging pattern classification of working memory operations (maintain, replace, suppress) shown as the area under the curve (AUC) where chance is 0.5. White dots in the violin plots are individual participants. The within-participant and between-participant classifications were performed on data from Bretton *et al.* (37), and between-experiment classification was performed by training on data from Kim *et al.* (16).

remaining in such states) (44,45). Again, in this case it will likely be useful to develop multivariate predictive models at the individual level to enhance their precision and sensitivity.

On the other hand, there are also strong advantages to developing group-based or population-level models fit on data across multiple participants. Moreover, these models are developed on one set of participants and then applied directly to new participants, obviating the need to collect additional training data. This approach increases generalizability because population-level models emphasize common features of mental states that are shared across individuals and often different task contexts. Therefore, they can lower the barrier to tracking mental states in newly collected datasets related to mindfulness.

The population-level approach is the strategy used most often when developing neuromarkers. For such applications, it becomes increasingly important that the predictive models have high sensitivity, specificity, and generalizability across individuals, variants of a construct (e.g., different kinds of pain, or craving elicited by different types of drugs), and populations. Population-level models trained on some individuals and tested in other individuals without adjusting the model become particularly advantageous in this setting. If the brain representation for a construct of interest is largely consistent across individuals, population-level models borrow strength and stability from the availability of more data across a wider set of physiological conditions and can actually perform better than individualized models trained on limited individual-person localizer data (31). Moreover, training a single model and testing it broadly reduces the problem of different models tracking different processes in different individuals and also allows the models to be validated or rejected—and their boundary conditions established—across diverse types of tasks and in diverse populations.

Group-level models may not generalize well to certain populations, however, and challenges may arise when trying to apply a group-level model to populations that vary strongly in

various demographic dimensions, such as age and socioeconomic status, and even variation in meditation skill level or the type of training experienced (e.g., Vipassana, Zen, Yoga). To tackle these issues, it is crucial to create population-level databases for mindfulness research that allow for coding and filtering of samples based on relevant factors. Recent efforts, such as the new ENIGMA (Enhancing Neuro Imaging Genetics through Meta Analysis)-Meditation consortium (46), are working toward this goal. This growing database of neuroimaging studies could enable not only meta-analyses and mega-analyses but also the development of population-level models mentioned earlier.

Even for strongly validated models that appear to generalize well across many populations and contexts, such as the NPS, recent findings have shown that certain components of multivariate brain patterns exhibit high levels of individual variability, such as those located in higher-order transmodal regions (medial and lateral regions of the prefrontal cortex) (47). Thus, more work is needed to explore when and how to select individualized versus population-level models. Another potential is to develop more powerful strategies, such as the use of hybrid approaches (e.g., models initialized to group-level neuromarkers but then updated through idiographic training and testing on the individual to optimize sensitivity) (36). Hybrid approaches could also be a powerful area for real-time neurofeedback methods (45,48,49). While these have not been used in mindfulness research yet, they have been applied in other areas, such as sustained attention (50). In those cases, multivariate decoding methods have been combined with real-time neurofeedback to enable personalized and adaptive training and testing. As a final point, recent developments suggest the potential of improved alignment of brain data across individuals based on shared function rather than purely anatomical constraints [e.g., hyperalignment (51)].

The future holds great promise for being able to build robust, generalizable multivariate predictive models for decoding of mental states associated with or influenced by

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mindfulness. Continued interdisciplinary research and methodological advancements will be crucial in overcoming challenges and optimizing model accuracy for diverse populations. By integrating individualized approaches with robust population-level insights, the field of contemplative neuroscience can make significant strides in decoding complex mental states and enhancing the understanding of mindfulness.

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